

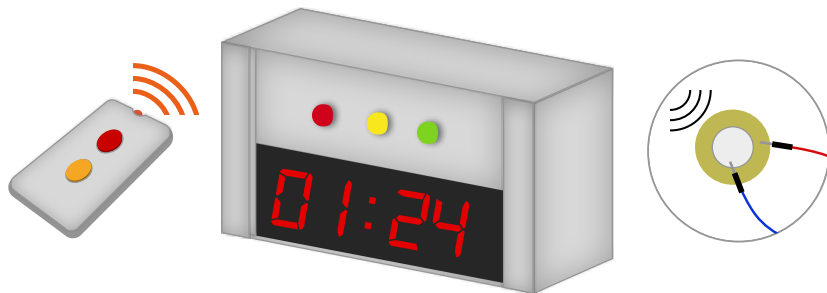
QUICKKNIFE: THE ALL-IN-ONE DEVICE FOR QUICK DRAW COMPETITIONS

The knife throwing community is a group of individuals who throw knives, axes, and tomahawks for sport. However, the scope of this project is only concerned with knives.

One set of knife throwing competitions is the quick draw, where two throwers must draw their knives and hit the target faster than the other. The concern is that in competitions with high skilled throwers, the knives hit the target within a fraction of a second from each other, making it impossible to tell by sound or sight who the winner is.

THE OBJECTIVES

1. Accurately record the duration of a knife throw, from when timer begins to when the knife hits the board.
2. Indicate the start of each round without causing disadvantages for players with various throwing preferences.
3. Build a strong and durable device that can handle knife hits as well as damage due to outdoor exposure.
4. Create a portable product that can easily be transported for competitions.



The proposed solution, the QuickKnife, addresses the needs of a quick draw competition.

THE FEATURES

1. **Piezoelectric sensor** — a sensor that converts mechanical deformation into electric charge, allowing it to determine knife strikes. It's cheap and easily accessible.
2. **LED screen and lights** — UV resistant, to ensure that nothing burns out in the sun.
3. **3 colored lights** — visual cues to indicate the beginning and end of each round. There's **green** for "Go", **yellow** for "Ready" and **red** for "Lose".
4. **One device for each target** — the player simply needs to look in the same direction they're throwing to see the indication light.
5. **Remote control** — for the referee to turn on the device or reset the time for each round.
6. **Polycarbonate frontal cover** — strong enough to withstand the blow of a knife without shattering.
7. **Stainless steel body** — designed to allow the frontal cover to easily slip out for a quick replacement, while also protecting the interior components.
8. **Plastic interior lining** — ensures that even if the circuit disconnects, it will not short with the metal body and damage the hardware.
9. Each device only weighs 1 kg and is approximately 16x10x10 cm.